

Conversation Provenance Recovery and Drift Audit

Executive finding

The strongest result of this audit is not that the historical record was wholly destroyed. It is that **the user-visible thread is not a sufficient provenance surface**.

A complete current-conversation file listing exposed **52 file records**: 43 generated artifacts and 9 uploads. All 52 were successfully materialized into this recovery bundle. The supposedly missing rebuild manual named in the Android notification was also located separately in Library as both Markdown and PDF. The recovered Markdown hashes to 0510b860982ba673d0f6e1da9a46fe9f583b5fc39ba4788c9da7de6c45b70731 and matches the SHA-256 printed in the recovered PDF/search record (0510b860982ba673d0f6e1da9a46fe9f583b5fc39ba4788c9da7de6c45b70731).

This changes the evidentiary conclusion. For the artifacts inspected here, there is strong evidence of **discoverability/location drift and representation drift**, but not evidence that the underlying Markdown content was silently rewritten. There is also clear **scope drift** across the conceptual documents. Whether each scope change was intentional development or assistant-induced misframing cannot be determined from artifact files alone and requires message-level comparison.

1. Recovery corpus

Current-conversation material recovered: **52 files**, totalling **8,775,730 bytes**.

File-type counts:

Type	Count
`.json`	1
`.md`	13
`.md.sha256`	4
`.pdf`	12
`.png`	20
`.txt`	2

Related Library recovery added: **2 files** (20260911_ANTHBLACK_SYSTEMS_REBUILD_MANUAL .md and . pdf). These were referenced by the earlier Android notification but were not part of the current-conversation file list.

2. Quantified preservation and duplication

Markdown

There are **13 Markdown file instances** representing **7 unique byte-identical Markdown states**. Therefore 6 of the Markdown instances are duplicate copies of already-present exact bytes.

The following repeated Markdown states were byte-identical:

- MILLENNIUM_ARGUMENT_DEVELOPMENT_RECORD: three copies, exact same SHA-256 c1f3ec53 . . .
- WRITING_DEVELOPMENT_METHOD_CONCEPT_DRAFT: three copies, exact same SHA-256 51209a17 . . .
- LOW_IDENTIFIABILITY_PROVENANCE_KEY_CONCEPT: two copies, exact same SHA-256 1051d7d9 . . .
- PROVENANCE_AUTHORITY_DRIFT_OBSERVATION_PENDING: two copies, exact same SHA-256 e913dc00 . . .

Measured Markdown content drift inside those duplicate groups: 0 bytes.

PDF

There are **12 PDF instances** and **12 distinct PDF binary hashes**. However, extracted-text comparison yields only **6 unique textual PDF states**.

That means multiple PDF files contain the same extracted text while differing at the binary level. Four conceptual records form repeated same-text/different-binary clusters:

- 20260911_LOW_IDENTIFIABILITY_PROVENANCE_KEY_CONCEPT.pdf (34,961 bytes, 852f0126afc9 . . .); 20260911_LOW_IDENTIFIABILITY_PROVENANCE_KEY_CONCEPT_2.pdf (34,961 bytes, 1171d44933cb . . .)
- 20260911_MILLENNIUM_ARGUMENT_DEVELOPMENT_RECORD.pdf (56,856 bytes, bac5beb074a7 . . .); 20260911_MILLENNIUM_ARGUMENT_DEVELOPMENT_RECORD_2.pdf (31,881 bytes, 48740e7eb295 . . .); 20260911_MILLENNIUM_ARGUMENT_DEVELOPMENT_RECORD_v0.1.0.pdf (56,856 bytes, ead72867969e . . .)
- 20260911_PROVENANCE_AUTHORITY_DRIFT_OBSERVATION_PENDING.pdf (31,847 bytes, 2b3561fc48bc . . .); 20260911_PROVENANCE_AUTHORITY_DRIFT_OBSERVATION_PENDING_2.pdf (31,847 bytes, 366edc3e9700 . . .)
- 20260911_WRITING_DEVELOPMENT_METHOD_CONCEPT_DRAFT.pdf (32,240 bytes, c8d4c1dd8ee6 . . .); 20260911_WRITING_DEVELOPMENT_METHOD_CONCEPT_DRAFT_2.pdf (7,270 bytes, 21f75ae3b7bf . . .); 20260911_WRITING_DEVELOPMENT_METHOD_CONCEPT_DRAFT_v0.1.0.pdf (32,240 bytes, b4caaf6de79c . . .)

This is **representation/serialization drift**, not demonstrated semantic drift. Different PDF bytes can arise from regenerated metadata, object ordering, font embedding, compression, timestamps, or renderer behaviour while preserving the visible/extracted text.

3. Scope drift across the document sequence

The unique Markdown corpus contains a clear sequence of changing scope frames:

Frame	Documentary scope	Classification
A	A specific public argument about AI, scientific humility, history and attribution	Topic-specific development
B	Writing development as a provenance-preserving process	Method abstraction
C	AI-assisted scientific reasoning and writing, with explicit testability/falsifiability	Broader methodological abstraction
D	Low-identifiability provenance and alias/key separation	Privacy/governance extension

Frame	Documentary scope	Classification
E	Potential provenance-authority/authorship drift	Meta-audit of the system itself

This produces **five materially different scope frames** and **four major scope transitions** in the preserved artifact sequence.

That number is not itself evidence of corruption. Some changes were clearly conceptual development. The important audit question is whether a scope transition was (a) initiated/accepted by the author, (b) introduced by the assistant and later corrected, or (c) introduced by later reconstruction without adequate provenance. The artifact set does not contain enough message-level evidence to classify every transition reliably.

4. Authorship drift

The recovered documents repeatedly contain explicit Human Gate and provenance language distinguishing author-origin material from assistant proposals or transformations. In the recovered artifact set, this audit found **zero directly demonstrated cases where a surviving original user-attributed idea is explicitly reassigned to the model in a later surviving document.**

Therefore the current evidentiary status is:

- **Observed risk:** later reconstruction can obscure or alter origin labels.
- **Directly demonstrated authorship inversion in this artifact corpus:** 0 confirmed instances.
- **Pending hypothesis:** at least one message-level case may exist and requires original-turn versus later-summary comparison.

This distinction matters. The documents support investigation of authorship drift, but they do not yet prove a specific authorship inversion by themselves.

5. Retrieval/location drift

The Android notification records that 20260911_ANTHBLACK_SYSTEMS_REBUILD_MANUAL.md and .pdf had been completed. Those files were not part of this current conversation's 52-file surface, but a targeted Library search recovered both artifacts.

The recovered Markdown SHA-256 is:

0510b860982ba673d0f6e1da9a46fe9f583b5fc39ba4788c9da7de6c45b70731

The recovered PDF record prints the same Markdown SHA-256:

0510b860982ba673d0f6e1da9a46fe9f583b5fc39ba4788c9da7de6c45b70731

This is strong evidence that the artifacts themselves persisted somewhere accessible to the account even though they were not visible where the user expected to find them. For this specific incident, **"missing from the thread" is not equivalent to "deleted from storage."**

6. Drift vector

A single percentage would overstate what can be measured. The defensible quantified result is a vector:

Dimension	Measured result	Evidentiary status
Current-conversation artifact recovery	52 / 52 listed file records materialized	Confirmed for file layer
Markdown duplicate content drift	0 bytes in 4 duplicate content groups	Confirmed
Markdown states	13 instances / 7 unique byte states	Confirmed
PDF binary stability	12 PDFs / 12 distinct binary hashes	Confirmed
PDF textual states	12 PDFs / 6 unique extracted-text states	Confirmed
Major preserved scope frames	5	Confirmed by document metadata/content
Major scope transitions	4	Confirmed sequence; cause mixed/uncertain
Confirmed authorship inversions	0 in recovered artifact corpus	Not demonstrated here
Explicit authorship-drift hypothesis records	1 pending record	Confirmed
Cross-surface location mismatch examined	1 incident involving 2 rebuild-manual artifacts	Confirmed
Rebuild-manual recovery	2 / 2 named artifacts recovered from Library	Confirmed

7. What this does and does not establish

Established

1. The current conversation has a larger recoverable artifact surface than the visible thread alone suggested.
2. Exact Markdown duplicates were preserved without byte drift.
3. Several PDFs were regenerated or serialized differently despite equivalent extracted text.
4. The conceptual scope changed materially across the sequence.
5. The rebuild manual referred to by the Android notification still exists in Library and its MD/PDF integrity relationship is internally consistent.

Not established

1. That ChatGPT routinely rewrites visible historical messages.
2. That conversation compaction deleted these files.
3. That any specific model or company intentionally changed authorship.
4. That all historical conversations and artifacts can be recovered from the currently available file surfaces.
5. That a scope change is an error merely because it is large.

8. Scientific interpretation

The evidence supports a narrower but technically important claim: **conversation continuity, artifact storage, UI visibility, memory/context synthesis, and provenance are separate layers and should not be treated as interchangeable**. A user-visible thread can fail as an archival interface even while the underlying artifact remains recoverable elsewhere. Binary regeneration can alter a file hash without altering its extracted text. Conceptual summaries can also change scope without any corresponding change to the original source artifact.

For defensible authorship research, the unit of evidence should therefore be the immutable original record plus timestamp, source identity, origin label, transformation link, and hash. Later assistant summaries should be treated as derived representations rather than authorities over origin.

9. Bundle structure

- `00_AUDIT_REPORT.md` - this report.
- `00_AUDIT_REPORT.pdf` - rendered report with exact Markdown SHA-256 in footer.
- `01_RECOVERY_MANIFEST.json` - hashes, sizes and source surface for every bundled recovered artifact.
- `02_SHA256SUMS.txt` - deterministic SHA-256 list for all recovered evidence files and report files.
- `current_conversation/` - all 52 files exposed by the current conversation file layer, including render images and duplicates.
- `recovered_library/` - rebuild manual Markdown and PDF recovered from Library.

10. Human Gate

This bundle is an evidence-preservation and preliminary audit artifact. Before using it publicly, legally, academically, or as evidence about platform behaviour, compare the relevant original message-level transcript/export against later summaries and classify each alleged drift event individually. Do not convert an observed discrepancy into a causal claim without that step.